

Supporting Early-Career Researchers: Strategies in Transdisciplinary Project Design

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Abstract. A growing body of academic literature highlights the inherent challenges in conducting transdisciplinary (TD) research. These challenges, particularly the institutional and cultural barriers, can be especially detrimental to those who are early in their academic career. To ensure the success of transdisciplinary efforts and prevent those involved from being disadvantaged, it is critical to develop practical actions to address these challenges. This paper presents a case study of the Made Smarter Innovation: Centre for People-Led Digitalisation, a £5 million, engineering led, transdisciplinary research centre. Drawing on prior experience of running TD projects, the Executive Team have implemented organisational and management strategies aimed at mitigating the negative impact on researchers. As the funding nears its conclusion, we conduct a focus group involving twelve Research Associates and Assistants, either currently or recently employed in the centre, to capture their reflections on these initiatives. The findings illustrate that organisational and management strategies can mitigate the challenges faced by researchers. Establishing a clear mission and vision for the project while giving researchers the freedom to design their studies and the journals they target, empowers them to manage their individual career path. Additionally, to alleviate the additional burden of industry outputs, it is essential to allocate funding for administrative and impact support within the project. Future efforts should focus on creating more structured career development programs which can be tailored to the career aspirations of the individual researcher.

Keywords. Transdisciplinary engineering, Transdisciplinary teams

Introduction

In an effort to address complex challenges, over the past five decades transdisciplinary (TD) approaches have received increasing attention. Indeed, within European policy, there is a suggestion that TD should be scaled up and become the “modus operandi” for research [1]. This support is reflected within the UK funding landscape where increasingly funding calls specifically request that a TD approach be taken [2-6].

Although receiving increased support, the literature shows that the nature of TD research – which requires the integration knowledge and methods from academic disciplines and non-academic partners – means that delivering a transdisciplinary project

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can be challenging [7-10]. A key issue which is identified is that TD research often takes longer to complete than traditional research and typically results in fewer publications or publications in less prestigious journals [11]. Given that a strong publication record is crucial for academic advancement, those early in their career may find working within a TD research project can hinder career progression [12, 13].

Despite recognising that working within a TD project may be unattractive for those who are early in their careers, few studies which have considered the practical actions which might be taken at project level to mitigate these issues. This is significant, taking the time to identify and reflect on actions that can support researchers in a TD environment will lead to the development of structures and strategies that could enhance their engagement and, ultimately, the success of transdisciplinary research.

Within this paper we present a case study of the Made Smarter Innovation: Centre for People-Led Digitalisation - a £5 million, engineering led, TD research centre. Aware of the potential negative impact for researchers, during the design of the project, organisation and management strategies were implemented aimed at mitigating the challenges. We present the findings of a focus group held with Research Associates and Assistants currently, or recently employed within the centre, where we elicit their reflections on these initiatives. These findings add to the body of knowledge and will be critical in informing future TD project design.

The paper is organised as follows: A review of the background literature is provided, focusing on the challenges of conducting TD research, with specific attention to the implications for those who are early in their career (Section 1). The centre and its organisation and management are described in detail (Section 2). The group discussion method used in this study is explained (Section 3). The results are then presented and discussed (Section 4). Finally, conclusions are drawn and areas for future research proposed (Section 5).

1. Background

Since gaining prominence in the 1990s, TD research has attracted increasing attention. Indeed, during the decade from 2014, there was a 300% rise in the number of papers referencing the term "transdisciplinary" in their title, keywords, or abstract on Scopus. Despite its growing popularity, the challenges of conducting TD research are well documented [7-10].

1.1. Classification of TD challenges

That there are numerous challenges in conducting TD research is well documented within the academic literature. For simplicity, a number of studies classify these challenges into categories. The following examples, illustrate some of the different approaches which have been taken.

In an early paper, Jakobsen, Hels [13] identify 30 different barriers and facilitators to TD research which they then classified into four hierarchical levels: individual-based, group-based, organisation culture-based, and a mixed category containing overlapping group and individual-based. In contrast, Gaziulusoy, Ryan [12] propose that challenges exist across four dimensions: inherent (which arise due to the nature of TD work), institutional, teamwork, emergent (changes in external factors). Meanwhile, Lang, Wiek

[14] mapped challenges across projects stages: framing the problem, creating the team, creating the solution, and then applying the co-created knowledge (Table 1).

Table 1. Classification of Transdisciplinary Challenges.

Jakobsen, Hels [13]	Gaziulusoy, Ryan [12]	Lang, Wiek [14]
4 Levels	4 Dimensions	4 Project stages
Individual-based	Inherent	Problem framing
Group-based	Institutional	Team creation
Organisation culture-based	Teamwork	Co-creation of the solution
Mixed (Group & Individual)	Emergent	Applying the co-created knowledge

1.2. Specific Challenges for Those Early in their Career

The literature highlights the multitude of challenges to TD research and that these challenges can have wide reaching implications for those early in their careers [15].

Co-creation and the development of socially relevant and implementable knowledge are essential components of TD research, requiring significant time and effort that should be properly acknowledged [16, 17]. However, projects often lack sufficient skills and resources to support public outreach and stakeholder engagement [11, 12]. These extra responsibilities, combined with the lower priority often placed on academic publications in TD research, reduce the time which can be directed towards academic journal publications [8, 12].

Aligned to this, academic performance criteria and evaluation systems often favour traditional disciplinary research, rather than TD research, which can be perceived as lacking rigor [9, 12, 13, 16, 17]. Those early in their career, feel more vulnerable in this regard compared to established academics with established publishing records [13].

Transdisciplinary research requires integrating theories from diverse disciplines [9]. Although new initiatives are starting to gain traction, there is a perceived lack of transdisciplinary networks [16]. This together with distributed teams, can hinder collaboration and make it challenging to foster the close-knit teamwork essential for successful TD working [12].

Despite the potentially negative impact that conducting TD research can have, there is a lack of guidance on how these challenges might be practically addressed. While some solutions emphasise institutional and educational reforms (e.g. Schrot [19], Rogga and Zscheischler [17], Witjes [18]), they often overlook practical strategies applicable within TD projects. Others propose concrete solutions (e.g. Jaeger-Erben [20], Djinlev [15]), but fail to assess their effectiveness. Our research contributes to the existing knowledge by presenting a case study of the Centre for People-Led Digitalisation, that evaluates practical TD project design strategies specifically designed to support researchers.

2. The Made Smarter Innovation: Centre for People-Led Digitalisation

The Made Smarter Innovation: Centre for People-Led Digitalisation (P-LD) is one of an ‘eco-system’ of initiatives funded as part of UK’s Made Smarter £300 million programme. The 3.5-year, £5 million project, aims to improve the outcomes of the adoption of digital technologies in manufacturing by encouraging a people-led approach whereby people and culture are given prior and explicit consideration during any digitalisation initiatives.

The project, which commenced in September 2021 is led by the Department of Mechanical Engineering at the University of Bath and encompasses academics and researchers from three Universities (Bath, Nottingham and Loughborough) from a range of diverse disciplines (engineering, philosophy, management etc) and non-academic partners from both the private, public and civil sectors.

2.1. Centre Organisation and Management Design

Recognising the challenges that undertaking a TD project can pose for a researcher's future success, it is crucial to develop strategies to address these issues. Within the P-LD, five specific organisational and management strategies were integrated into the project design.

2.1.1. Centre Leadership

In this challenging environment it is proposed that TD research is strengthened by having experienced leadership [21]. The P-LD is led by an academic from engineering who works closely with the Executive Team which comprises of the academic leads from the two other Universities, a Research and Industry Engagement Manager, and a Centre Manager. This group have extensive experience of managing transdisciplinary research projects.

Guidance from supervisors is considered to play a key role in overcoming TD challenges [16]. As such, in addition to the Executive Team, each researcher is allocated an academic line manager; in some cases, both are from the same discipline, whilst in others, they come from different disciplines. This blending of disciplines was an intentional attempt at creating a dynamic environment that fosters cross-disciplinary exchange.

2.1.2. Research Cohort Approach

A study by Djinlev [15] concludes that key to supporting researchers is creating a space for informal exchange where concerns and questions can be raised without fear that their questions may be trivial. Within the P-LD the research team are not co-located within a single office or university building. To encourage exchange across the disciplinary boundaries a cohort approach is taken. Under the direction of a dedicated Research and Industry Engagement Manager, all researchers and PhD students come together on-line, weekly, to share information and knowledge. Outside of these on-line meetings, ad-hoc in-person social events and training are organised.

2.1.3. Research and Outputs

According to the seminal work of Jantsch [22], TD methods should not reside within a single discipline but should evolve to reflect the context in which the work is being conducted. Embracing this, the P-LD leadership adopts a relaxed approach to research management. Provided that the research delivers high quality research meeting the end goal of the project, researchers are empowered to decide the focus, the approach taken, and the outputs they deliver. In terms of academic publications, the researchers are steered towards high-ranking journals but given the freedom to publish in whichever journal they consider most appropriate and relevant for their future career.

2.1.4. Administrative and Impact Support

Transdisciplinary research has an increasing demand for non-academic engagement [16]. A key measure of the success of the centre is the delivery of tools and insights which are practical and useful for industry. This creates additional workload and often entails undertaking tasks which are beyond the skills and expertise of researcher. Within P-LD, funding was allocated to dedicated support for administration and impact activities. Examples of where support has been provided includes booking travel to events, making industry connections, workshop organisation, and assistance in writing industry focussed articles.

2.1.5. Skills and Career Development

TD research requires skills beyond those often taught within traditional disciplinary education [15]. The project budget included funding for activities aimed at enhancing researcher skills and career development opportunities. The support extended to a range of activities designed to strengthen both academic and industry skills, such as funding for conference travel, courses in Machine Learning, blog writing training, and professional careers coaching.

3. Research Approach

The organisational and management strategies deployed within the P-LD were informed by the Executive Team based on their prior experience in conducting TD research. Although well-intentioned, to evaluate their effectiveness and impact it is crucial to understand how these initiatives were perceived by the researchers themselves. Towards this, we conducted a group discussion with Research Associates and Assistants currently or recently employed within the centre. The aim was to explore their experiences regarding the Centre's organisation and management – what they considered both positive and negative.

To encourage the participants to share their thoughts, ideas, and experiences freely the session was organised and facilitated by an independent person unaffiliated with the Centre, and took an unstructured, exploratory approach. The responsibilities of the facilitator included recruiting participants, ensuring informed consent was obtained, and facilitating the discussion.

The session was held on Microsoft Teams, lasted 90 minutes and involved twelve participants (P), representing 100% of the eligible pool. The discussion was automatically transcribed by Teams, with the facilitator anonymising the data before transferring it to the study lead for thematic analysis. To ensure a systematic and rigorous analysis, the thematic analysis followed the process outlined in the work of Braun and Clarke [23], and was supported by NVivo computer-based software.

4. Results and Discussion

The group discussion involved twelve Research Associates and Assistants from ten different disciplinary backgrounds: Computer Science, Education, Economics, Economics & Management, Information Systems, Management (x 3), Manufacturing,

Mechanical Engineering, Philosophy, and Policy. In conducting the analysis, the statements were initially coded against the five operational and management strategies outlined in Section 2.1: centre leadership (2.1.1), research cohort (2.1.2), research outputs (2.1.3), administrative and impact support (2.1.4), and skills and career development (2.1.5) and within these, sub-categorised to identify emergent themes.

4.1. Centre Leadership

Reflections from the group with regards to the centre's leadership were positive. It was recognised that the leadership showed a "remarkable level of dexterity in managing a diverse team" and were able to "keep them [researchers] satisfied and engaged" (P5). Beyond leadership, the participants appreciated the role that leadership played in "integrating all the information from the cohort's people" (P7).

The comments supported that good communication was occurring with clear communication of expectations from the leadership (P11 & P6). They reported a feeling that they could ask questions without fear "I felt like I was always able to ask a question, even if it was something that I thought was a dumb question or that I was sure I'd already been told" (P2). Consequently, the environment was considered "less stressful" (P2). The positive feedback regarding leadership is consistent with the literature, where experienced leadership is considered important in managing the diverse demands of TD research teams [21].

Reflecting on the strategy of having a Researcher managed by someone outside of their discipline there was a mixed response. There was an understanding of the reasoning behind this approach. However, they identified that there could be potential benefits if their research aligned more closely with that of their line manager. They suggested that a possible solution could be to have two line managers - one from their own discipline and another from a different discipline (P2). The literature suggests that TD networks are an important element of TD success [16]. An important insight is that although creating the environment for TD exchanges may encourage collaboration, interdisciplinary supervision may result in insufficient disciplinary support.

4.2. Research Cohort Approach

The cohort approach comprises of two elements – an on-line, weekly meeting and ad-hoc in-person events such as training and socials. The researchers generally agreed that the on-line meetings were informative. The participants appreciated that they were "very efficient and straight to the point" (P6) and they were a good way for the cohort to be "kept in the loop" (P2). The on-line format was considered "a very good idea" as people did not have to attend in person (P3).

Although several comments supported that they were a good means through which to keep the cohort informed, their success as a means to breakdown silos might be questioned as one person commented that "There are some people who I've still got no idea what they're actually researching" (P2).

Most of the team (ten participants) are based in Bath, with only one researcher at each of the other two locations. This introduced concerns that there was a core of people (those at Bath), and a peripheral group. Holding training events and meetings in different locations was seen as a potential solution.

One concern not specifically related to the cohort approach but rather the feeling of the group as a collective was that the priority was given to the industry. Comments made

included “I feel like I’m not treated as well as an industry partner...events that were exclusive for industry partners...I thought could actually be helpful for us” (P9). There was a feeling of being excluded from industry social or planning activities “We are all invited to most of the event, but then at the end, they could leave with the industry partners to go to dinner” and “afterwards there might be a second part of [an] event that is only for industry partners” (P9). Although expressing that this might be understandable as “they have other things to discuss”, they considered that it might be explained or handled differently.

The concern about silos within the cohort and a peripheral group highlights a common challenge in TD research where distributed teams struggle to collaborate effectively due to geographical separation [16]. This is compounded by feelings of exclusion from industry-specific activities, underscoring the literature’s assertion that those working in TD projects often feel marginalised [12].

4.3. Research and Outputs

P-LD is managed in a way that allows researchers to focus on their own research interests, as long as their work aligns with the centre’s overall vision and mission. The participants were appreciative of this freedom and the openness of the leadership team to their research ideas “that freedom to research what you want in the way you want it and write it up and send it off to the publications where you think it’s best and so on, that for me has been the best feature of the whole thing” (P2).

Somewhat surprisingly, although challenges in publishing is often seen as a major challenge [9, 12, 13, 16], only one comment mentioned this as an issue “getting the right journal and getting them to accept your paper for any transdisciplinary project is very difficult” (P4). This perhaps reflects the empowerment of the researchers to define their own research studies and journal selection.

Of greater concern was the changing priorities and how these impacted publications “So they wanted us to make a sudden shift from what we are doing. So, it definitely affected the publication timeline for instance.” (P3). Despite enjoying their research freedom, concerns were raised around unclear expectations, timelines for outputs and changing priorities. Although one participant reflected that they liked the lack of deadlines and that having “a particular number of papers or something, then I think that it would have increased the pressure” (P2), some wanted clearer targets for outputs (P8 & P5).

Last minutes changes such as money becoming available for conferences and an industry event being brought forward, introduced stress “they told us you have to register for a conference. I was like, OK. I need time to think.” (P3), “This is the time you have to book your hotel...OK, we can do it, but not in that hurry. We need more time.” (P3), “suddenly they decided industry is more important and they need a tool to develop so the industry can use it... I really enjoyed the output, but the way it happened it was just so sudden” (P3).

4.4. Admin and Impact Support

Comments with regard to support with administration and impact activities were entirely positive showing an appreciation of the diversity and generosity of support provided “she’s always like really, really super...highlighting events that might be relevant...and is very on it when you needed help with expenses or travel” (P1), “So it was always very

easy to contact her...whenever we have a problem or whenever we need some clarification. She was always there for us.” (P3). This was seen as beneficial “it allows [us] to do [our] research without being really worried or pressed with different tasks” (P8), “aside from having an academic managing that has been really valuable and then having that admin support and that access really has made a huge difference and you can see that difference in all of the comments that everybody's making.” (P12).

Such positive feedback regarding support aligns with the literature's call for sufficient support structures to mitigate the additional administrative burden [15]. Effective administrative support allows researchers to focus more on research and less on logistical concerns, which is essential in alleviating the stress often associated with TD projects.

4.5. Skills and Career Development

The participants appreciated that training had been proactively arranged on their behalf “I think they were just always very proactive with looking out for stuff” (P1). They reflected that they had done “quite a lot of training sessions” (P1) as a group and that these sessions had been “relevant to all of us” (P1). Comments were made that they felt “really supported” (P6) to improve themselves “I think one of the best things in P-LD for me is [that] they gave us a lot of growth opportunities, the support our personal development, our personal growth” (P10). The opportunity to undertake training was considered “a positive thing” (P9) and “maybe this is something that we don't see much, [when] comparing with other departments” (P9).

Through their associated with the project, they had learnt to use “the language of industry partners” (P9) and “how to really write in the industry context” (P11) but accessing teaching opportunities has been challenging which may hinder their ability to progress in an academic career “there was nothing like teaching opportunities, to help me get a full-time academic position in future” (P11). Consequently, it was reflected that “there was more focus on somebody to develop industry related career. But for somebody who's developing [as an] academic there wasn't any emphasis on developing [this]” (P11).

This was compounded by the lack of clarity around the financial support available for training, with a reflection that “So if we knew that, what are the options, we could be more proactive because we didn't know how we could spend the centre's money.” (P3).

The participants' appreciation for training opportunities highlights the positive impact of skill development within a TD framework. However, the lack of academic career development opportunities, such as teaching experience, resonates with the literature's concern that TD research often prioritizes industry-related career development over academic progression [15]. Researchers in TD projects may face challenges in building an academic portfolio, which is essential for securing long-term academic positions.

5. Conclusion

The literature shows there are numerous challenges to conducting TD research and that these challenges disproportionately impact those who are early in their career. Within this work we explore whether strategic choices around organisation and management have the potential to create environments which are more attractive and supportive to researchers.

The results indicate that the strategic operational and management design of P-LD has helped mitigate some common challenges typically experienced by researchers. The difficulty in achieving academic publications, commonly highlighted as a challenge in the literature, does not present as a major concern for the participants within our study.

The leadership team's experience in managing TD research and the research cohort approach, which includes weekly online meetings, appears to have contributed to positive and open communication within the project. However, the success of this approach in fostering collaboration across disciplines is less clear. One significant finding is that forcing interdisciplinary interactions - such as assigning line managers from outside a researcher's discipline - can have unintended consequences.

The administrative and impact support provided to researchers within the centre has shown to reduce the additional burdens on researchers, allowing them to focus on their work while developing new skills in a supportive environment. We propose that in designing TD projects ensuring that this support is provided is critical.

Whilst there is an appreciation of training to support the researchers in their career development, implementing more structured career development programs which account for the needs of those intending to remain in academia, would be beneficial.

It is important to note, that these conclusions are based on findings from a specific group and context. As such, they may not be generalisable to all TD research environments. Further research is needed to test whether these insights hold true across other projects and institutional settings.

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References

- [1] Organisation for Economic Co-operation and Development, *Addressing societal challenges using transdisciplinary research*, OECD Global Science Forum, Editor. 2020, OECD., p. 80.
- [2] UKRI. *Land use for net zero – research (LUNZ-Research)*. 2023 [cited 2025 21 March]; Available from: <https://www.ukri.org/opportunity/land-use-for-net-zero-research-lunz-research/>.
- [3] UKRI. *Maximising UK adaptation to climate change hub*. 2023 [cited 2025 21 March]; Available from: <https://www.ukri.org/opportunity/maximising-uk-adaptation-to-climate-change-hub/>.
- [4] UKRI. *Resilient UK coastal communities and seas*. 2023 [cited 2025 21 March]; Available from: <https://www.ukri.org/opportunity/resilient-uk-coastal-communities-and-seas/>.
- [5] UKRI. *Maximising UK adaptation to climate change research projects*. 2024 [cited 2025 21 March]; Available from: <https://www.ukri.org/opportunity/maximising-uk-adaptation-to-climate-change-research-projects/>.
- [6] UKRI. *UKRI systems approaches to advance the UK's transition to net zero: full*. 2024 [cited 2025 21 March]; Available from: <https://www.ukri.org/opportunity/ukri-systems-approaches-to-advance-the-uks-transition-to-net-zero/>.
- [7] M.G. Lawrence, et al., *Characteristics, potentials, and challenges of transdisciplinary research*. One earth., 2022, 5(1), pp. 44-61.
- [8] N. Wognum, W. Verhagen, J. Stjepandic, Trans-disciplinary systems as complex systems, *Advances in Transdisciplinary Engineering*, 2017, Vol. 5, pp. 745–754.

- [9] K. Thapa, W.J.V. Vermeulen, and P. Deutz, Science with society: Challenges of early - stage researchers engaging with transdisciplinary research in sustainability science. *Sustainable development*, 2022, 30(6), pp. 1562-1572.
- [10] S. Lattanzio, Y.M. Goh, R. Houghton, L. Newnes, The Challenges of Conducting Transdisciplinary Engineering Research: A Case Study of the Made Smarter Innovation: Centre for People-Led Digitalisation. *Advances in Transdisciplinary Engineering*, 2023, 41, pp. 751–760.
- [11] M. Strand, et al., Transdisciplinarity in transformative ocean governance research – reflections of early career researchers. *ICES Journal of Marine Science*, 2022, 79(8), pp. 2163–2177.
- [12] A.I. Gaziulusoy, et al., Identifying and addressing challenges faced by transdisciplinary research teams in climate change research. *Journal of Cleaner Production*, 2016, 123, pp. 55-64.
- [13] C.H. Jakobsen, T. Hels, and W.J. McLaughlin, Barriers and facilitators to integration among scientists in transdisciplinary landscape analyses: a cross-country comparison. *Forest policy and economics*, 2004, 6(1), pp. 15-31.
- [14] D.J. Lang, et al., Transdisciplinary research in sustainability science: practice, principles, and challenges. *Sustainability science*, 2012, 7(SUPPL. 1), pp. 25-43.
- [15] V.D. Djinev, et al., Challenges and strategies in transdisciplinary research – early career researchers’ perspectives. *GAIA - Ecological Perspectives for Science and Society*, 2023, 32(1), pp. 172-177.
- [16] E.A. Nyboer, et al., Goals, challenges, and next steps in transdisciplinary fisheries research: perspectives and experiences from early-career researchers. *Reviews in fish biology and fisheries*, 2023, 33(2), pp. 349-374.
- [17] S. Rogga, & J. Zscheischler, Opportunities, balancing acts, and challenges - doing PhDs in transdisciplinary research projects. *Environmental Science & Policy*, 2021, 120, pp. 138-144.
- [18] S. Witjes, & Vermeulen, W. J. V., Transdisciplinary research: Approaches and methodological principles. in M.M. Keitsch et al. (eds.) *Transdisciplinarity For Sustainability - aligning diverse practices*. Routledge. London, 2021, pp. 27–52.
- [19] O.G. Schrot, Krimm, H., & Schinko, T, Enabling early career sustainability researchers to conduct transdisciplinary research: Insights from Austria. *Challenges in Sustainability*, 2020, 8(1), pp. 30–42.
- [20] M.K. Jaeger-Erben, et al., Building Capacities for Transdisciplinary Research: Challenges and Recommendations for Early-Career Researchers. *GAIA - Ecological Perspectives for Science and Society*, 2018, 27(4), pp. 379-386.
- [21] P. Angelstam, et al., Solving Problems in Social–Ecological Systems: Definition, Practice and Barriers of Transdisciplinary Research. *Ambio : a journal of the human environment*, 2013, 42(2), pp. 254-265.
- [22] CERI, *Interdisciplinarity: Problems of teaching and research in Universities*. 1972.
- [23] V. Braun, & V. Clarke, Using thematic analysis in psychology. *Qualitative Research in Psychology*, 2006, 3(2), pp. 77-101.